



Ankit Kumar

Roll No.: BTECH/10097/22

Electronics and Communication

Bachelor of Technology (B.Tech.)

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🔗 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

- **Birla Institute of Technology, Mesra** 2026
B.Tech in Electronics and Communication Engineering CGPA: 8.25
- **Narbheram Hansraj English School** 2022
Indian School Certificate (ISC), New Delhi Percentage: 95
- **Hill Top School** 2020
Indian Certificate of Secondary Education (ICSE), New Delhi Percentage: 94.2

EXPERIENCE

- **Samsung Research & Development, Noida** Jan 2026 - Present
Assistant Engineer Noida
 - Developed an internal automation tool to parse and extract valid CA combos from official Qualcomm documentation, cutting lookup time from 15+ mins to seconds. Support for multi-combo intersection queries and sortable, searchable interface.
 - Developed a backend utility using Android Debug Bridge (ADB) for device communication, log retrieval, detection and flagging of illegal IMEI instances.
 - Designed an intuitive front-end interface for the ADB utility, enabling non-technical customer service staff to independently perform IMEI validation without requiring engineering support.
- **Tata Motors** May 2025 - July 2025
Software Engineering Intern Jamshedpur
 - Developed a real-time safety compliance system using YOLOv10 and Python on live CCTV feeds, achieving 94%+ accuracy in detecting helmet violations and concurrently tracking engine count along the production line.
 - Engineered a fault-tolerant automated data pipeline over a local network to aggregate Excel reports from 11 edge systems to a central surveillance node at regular intervals, with robust error handling for network and system failures.

PERSONAL PROJECTS

- **Inventory Management Microservices Backend** 🔄 August 2025
Tools & technologies : Spring Boot, MongoDB, OpenFeign, REST API, Kafka, Java MailSender, Docker
 - Designed a scalable microservices-based backend system having independent Product, Supplier, Order, Notification and Inventory services, supporting CRUD operations for each service and real-time inventory management through RESTful endpoints and centralized API Gateway.
 - Used inter-service communication with OpenFeign clients and MongoDB for persistent storage, ensuring automated handling of inventory records. Used Kafka asynchronous communication between order and notification service.
 - Containerized services using Docker and implemented CI/CD pipelines using GitHub Actions for automated build and deployment.
- **QueryInsight : AI-Powered SQL Chatbot & Visualization Dashboard** 🔄 July 2025
Tools & technologies: Python, LangChain, OpenAI API, SQLAlchemy, MySQL, and Plotly/Dash.
 - Developed a robust pipeline that converts complex natural language questions including those requiring multi-table JOINS and aggregations—into executable MySQL queries, with database schema introspection and advanced result validation.
 - Integrated an interactive data visualization module using Plotly/Dash, enabling dynamic charting for query results, with comprehensive safeguards for SQL safety and semantic accuracy.
- **Predictive Maintenance (RUL Prediction)** 🔄 February 2025
Tools & technologies: Deep Learning, TensorFlow, Random Forest, Scikit-learn
 - Built a predictive maintenance system to estimate Remaining Useful Life (RUL) of aircraft engines using NASA's C-MAPSS dataset, implementing multiple regression and classification models, including SVR, Random Forest, and XGBoost.
 - Achieved optimal performance through extensive feature engineering, data preprocessing, and cross-validation.

TECHNICAL SKILLS

Backend Development: Java, Spring Boot, REST APIs, Microservices Architecture

Databases: MongoDB (NoSQL), MySQL

Tools & Technologies: Kafka, Docker, Git, GitHub Actions (CI/CD)

Networking: TCP/IP, OSI Model, HTTP/HTTPS, Network Programming

Frontend: Dash, Streamlit (Data Visualization UI)

Machine Learning: Python, TensorFlow, LangChain, Computer Vision (YOLO), NLP

ACHIEVEMENTS

- **Graduate Aptitude Test in Engineering (GATE) in DA** Secured AIR 1388 February 2025
- **IIT Bombay Techfest (Robowars)** Secured a position among the top teams worldwide. December 2024